

Yuhan (Hailey) Liu

PhD Candidate in Applied Economics

New York University Abu Dhabi and Southwestern University of Finance and Economics

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RESEARCH PROFILE

Fields: Urban and transport economics; policy evaluation; environmental and energy economics; empirical industrial organization; discrete choice model

I study how urban regulation, infrastructure, and market incentives shape technology adoption, resource allocation, and social welfare. My methodology combines causal inference, structural demand estimation, discrete choice models, counterfactual policy analysis, and welfare analysis.

EDUCATION

Southwestern University of Finance and Economics (SWUFE), China 2022-2027

Direct PhD Program in Applied Economics, Institute of Western China Economic Research

Dissertation research: The Social Welfare Effects of China's Electric Vehicle Industry from Macro, Meso, and Micro Perspectives

New York University Abu Dhabi, UAE 2025-2026

Jointly Supervised PhD Program in Civil and Urban Engineering

Southwestern University of Finance and Economics, China 2018-2022

Bachelor of Economics

ACADEMIC RESEARCH

Liu, Y., Hu, Y., and Zheng, X. (2025). **Effectiveness and Efficiency of Incentivizing Policies for New Energy Vehicles: From the Perspective of Powertrain Type**. Transportation Research Part A: Policy and Practice 199, 104530. doi:10.1016/j.tra.2025.104530

Structural demand estimation and counterfactual welfare analysis using 850,306 municipality-month-model observations. Current incentives raised adoption but generated a modeled net welfare loss; efficient targeting differs by vehicle powertrain.

Liu, Y., Zheng, X., Mao, Z., and Hu, Y. **Price or Quantity? Second-Best Electric Vehicle Promotion under License-Plate Rationing**. Under review at the Journal of Development Economics. (My job market paper)

Combines triple differences, event-study evidence, and a random-coefficients demand model to distinguish price relief from access relief under a binding urban quota. A within-quota subsidy produces less EV adoption but greater welfare than a quota exemption because it limits market expansion and external costs.

Liu, Y., Zheng, X., and Cherchi, E. **Price War, Perceived Quality, and Welfare: A Hybrid Choice Model of Electric Vehicle Demand**. Revise and Resubmit at Transportation Research Part A: Policy and Practice.

Estimates a control function-hybrid choice model using 45,064 purchases. The price war increase utility through lower purchase costs but are also negatively associated with perceived quality, with a stronger negative coefficient than the positive coefficient on price premiums.

Zheng, X., Liu, Y., Ni, E., and Nepal, R. **Impacts of the Belt and Road Initiative on the Energy Efficiency of Related Countries**. Under review at Economic Change and Restructuring.

Uses stochastic frontier analysis, staggered difference-in-differences, and event studies for 149 countries from 1995-2023. Belt and Road participation improves energy efficiency by approximately 1.9 percentage points through renewable-energy transition and stronger efficiency governance.

Liu, Y. and Cherchi, E. **Synthesizing and Monetizing the Social Welfare of Electric Vehicles: An Integrative Review across Transportation, Energy, and Economics**. Working paper.

Develops a reproducible five-account welfare framework spanning vehicle markets, public finance, environment and health, network integration, and dynamic industrial and innovation effects.

POLICY AND PUBLIC ENGAGEMENT

Policy Advisory Report, Coauthor

2025

- Coauthored an evidence-based report on electric-vehicle industry standardization that was adopted by a provincial chief in China.

Development and Reform Bureau, Secondment

2025

- Consolidated economic and social data and contributed to the government's mid-year performance report and planning work for the Fifteenth Five-Year Plan during a field-based policy secondment.
- Conducted village and enterprise fieldwork and developed a combined digital-marketing and retail-partnership proposal for the local industry; the work was covered by official media in China.

METHODOLOGY AND SOFTWARE

- Structural and behavioral methods: random-coefficients demand, nested logit, multinomial and mixed logit, hybrid choice models, latent-variable measurement, sampled choice sets, and Gauss-Hermite quadrature.
- Causal and policy evaluation: difference-in-differences, triple differences, staggered adoption, event studies, instrumental variables, matching, fixed effects, robustness and sensitivity analysis, and stochastic frontier analysis.
- Data and computation: Python, MATLAB, Stata, SPSS, LaTeX, and Microsoft Office; experience integrating structured and unstructured sources and applying multiple imputation and price imputation.

FELLOWSHIP, FUNDING AND AWARD

National Graduate Student Fellowship, China Scholarship Council, US\$24,000

2025-2026

Merit Fellowship, SWUFE, RMB 25,000 per year

2022-2027

Field Study Fellowship, SWUFE, RMB 30,000

2024

Teaching Fellowship, SWUFE, RMB 10,000

2023-2024

Second Prize in the Industrial Economics Paper Competition

2024

CONFERENCE PRESENTATION

Second Annual Conference of the International Society for Energy Transition Studies

2024

Impacts of the Belt and Road Initiative on the Energy Efficiency of Related Countries.

China Young Economist Society Summer School On Real-World Economic Research

2024

TEACHING EXPERIENCE

Teaching assistant for the undergraduate course "Mathematical Modeling"

2024

Teaching assistant for the undergraduate course "Microeconomics"

2023